

Lab 3 Prelab Assignment: PID Compensation for DC Motor

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1 Plant Definition

The plant used in this experiment is based on the velocity transfer function obtained from Experiment 2. To allow for a stability limit, an additional lag term is introduced:

$$H_{\text{lag}}(s) = \frac{1}{1 + \frac{s}{40}}$$

Thus, the plant becomes:

$$G_p(s) = \frac{302.5}{s(s + 35.09) \left(1 + \frac{s}{40}\right)}$$

Rewriting in standard polynomial form:

$$G_p(s) = \frac{12100}{s(s + 35.09)(s + 40)} = \frac{12100}{s^3 + 75.09s^2 + 1404s}$$

2 Task 1: Ziegler-Nichols Tuning Simulation

2.1 Objective

Determine initial PID parameters using the **Ziegler-Nichols Ultimate Gain Method**.

2.2 Method

A proportional controller $C(s) = K$ is applied. MATLAB is used to determine the critical gain K_c (gain at sustained oscillations) and the critical period T_c .

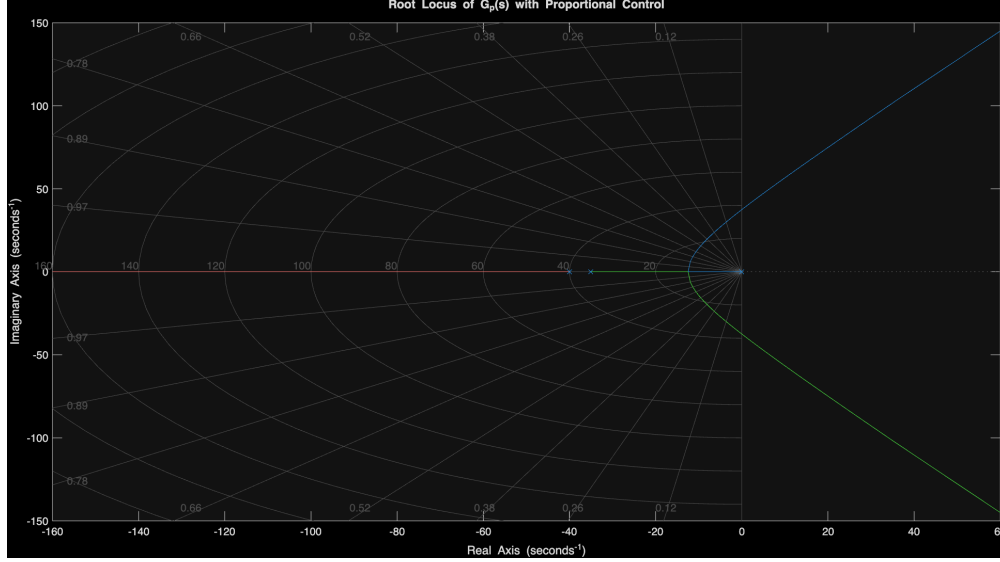


Figure 1: Root Locus of the P Controller

2.3 Results

- **Critical Gain:** $K_c = 8.7104$
- **Critical Period:** $T_c = 0.7642$ s

2.4 PID Parameters (Ziegler-Nichols)

Using the ZN tuning rules:

- $K_p = 0.6K_c = 5.2263$
- $T_i = \frac{T_c}{2} = 0.3821$ s
- $T_d = \frac{T_c}{8} = 0.0955$ s
- $K_i = \frac{K_p}{T_i} = 13.6771$
- $K_d = K_p T_d = 0.4993$

3 Task 2: PID System Analysis & Root Locus

3.1 Closed-Loop System

The PID controller is defined as:

$$C(s) = K_p + \frac{K_i}{s} + K_d s$$

The closed-loop transfer function is:

$$T(s) = \frac{C(s)G_p(s)}{1 + C(s)G_p(s)}$$

3.2 Step Response Analysis

The closed-loop system was simulated using a unit step input.

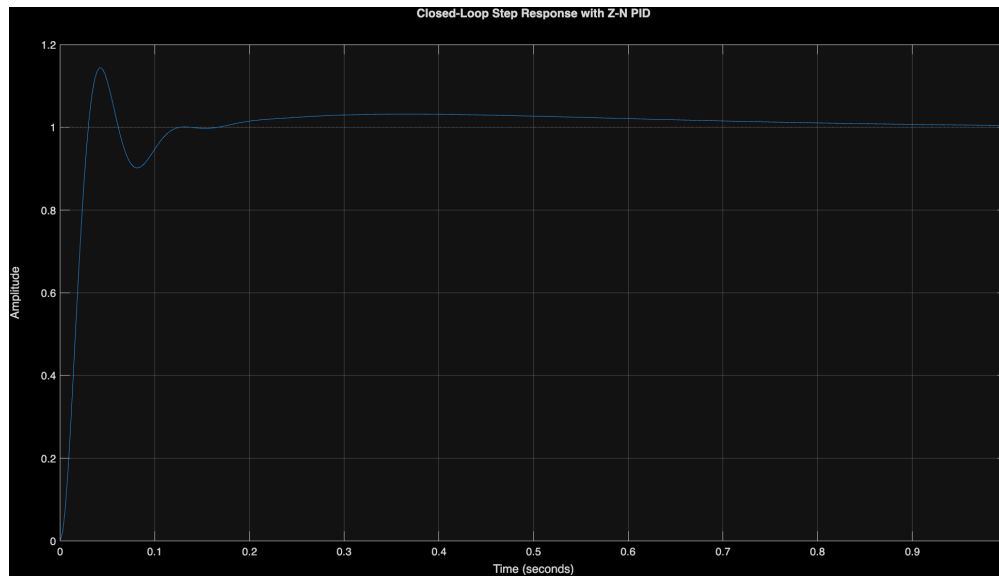


Figure 2: Closed-Loop PID Step Response

Observations:

- The system is stable.
- The response is underdamped with decaying oscillations.
- The controller improves response speed compared to P-only control.

3.3 Closed-Loop Poles and Zeros

- **Poles:** $-33.0287 \pm 75.7157i$, $-4.5163 \pm 1.9636i$
- **Zeros:** -5.2340

3.4 Root Locus (Varying K , Fixed T_i, T_d)

The controller structure for the locus is: $C(s) = K \left(1 + \frac{1}{T_i s} + T_d s \right)$.

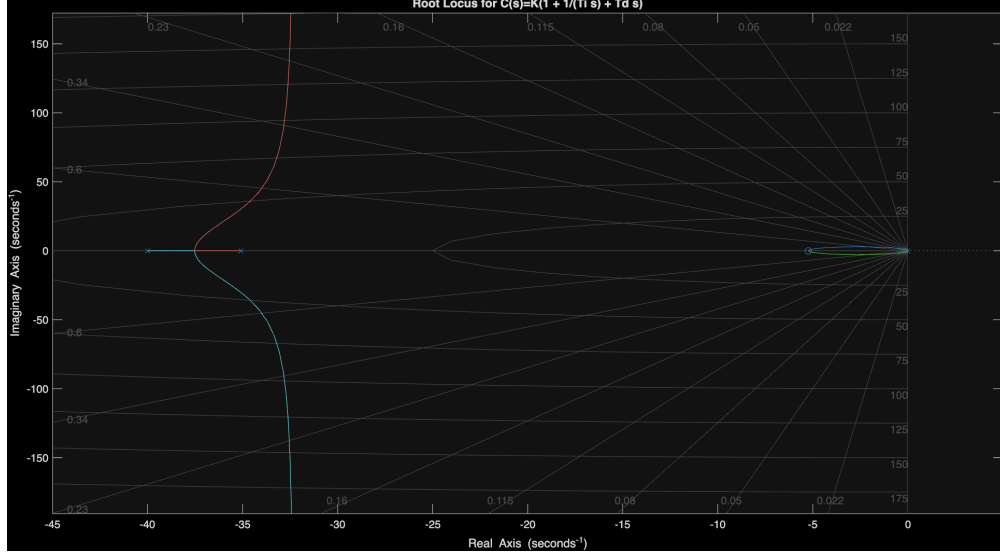


Figure 3: Root Locus of the PID Compensated System

4 Task 3: Derivative Term Discretization

We aim to approximate $d(k) \approx \frac{de(t)}{dt}$.

4.1 Forward Rectangular Approximation

Using $s \approx \frac{z-1}{T}$:

$$d(k) = \frac{e(k) - e(k-1)}{T} \quad (\text{Causal implementation})$$

4.2 Backward Rectangular Approximation

Using $s \approx \frac{z-1}{Tz}$:

$$d(k) = \frac{e(k) - e(k-1)}{T}$$

4.3 Bilinear (Tustin) Transformation

Using $s \approx \frac{2}{T} \frac{z-1}{z+1}$:

$$d(k) = -d(k-1) + \frac{2}{T}[e(k) - e(k-1)]$$

5 Task 4: Derivative Filter Design

5.1 Filter Specifications

- **DC Gain:** 1
- **Cutoff Frequency:** 200 Hz $\implies \omega_c = 400\pi \approx 1256.64$ rad/s

5.2 Continuous-Time Transfer Function

$$H(s) = \frac{\omega_c}{s + \omega_c}$$

5.3 Discretization

1. **Forward Euler:** $y(k) = (1 - \omega_c T)y(k-1) + \omega_c T x(k-1)$
2. **Backward Euler:** $y(k) = \frac{1}{1+\omega_c T}y(k-1) + \frac{\omega_c T}{1+\omega_c T}x(k)$
3. **Bilinear (Tustin):** $y(k) = \frac{2-\omega_c T}{2+\omega_c T}y(k-1) + \frac{\omega_c T}{2+\omega_c T}x(k) + \frac{\omega_c T}{2+\omega_c T}x(k-1)$

$$H_c(s) = \frac{1257}{s + 1257}$$

Forward Euler: $y(k) = -0.256637y(k-1) + 1.256637x(k-1)$

Backward Euler: $y(k) = 0.443137y(k-1) + 0.556863x(k)$

Tustin: $y(k) = 0.228261y(k-1) + 0.385870x(k) + 0.385870x(k-1)$